Standard Bank Report

1. Customer Summary

The customer exhibits a mixed financial profile. Strong cash inflows are evidenced by sizable salary credits in specific months, yet the monthly surplus is highly volatile—with several months registering deficits that undermine the ability to consistently meet fixed EMI obligations. While the average monthly surplus is around 855.60, the inconsistency in cash flow suggests the need for a lower, more conservative EMI threshold (recommended at approximately 733 per month). Furthermore, the borrower maintains disciplined spending habits and benefits from a very low overall DTI of 5.37%, supporting a sound long-term income base.

Surplus Position

Below is our detailed analysis using only the provided tables:

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1. Q1: Consistency to Cover an EMI Amount

- Assumption: In this analysis we take the “emi_amount” to be equal to the customer’s average monthly surplus of approximately 855.60.
- Using the Monthly Surplus Table, we compare each month’s surplus to 855.60:
  – May 2023: 7170.56 (well above)
  – June 2023: –131.66 (below)
  – July 2023: –1427.84 (below)
  – August 2023: –2221.87 (below)
  – September 2023: –946.17 (below)
  – October 2023: 1466.27 (above)
  – November 2023: –2238.29 (below)
  – December 2023: 5173.79 (above)

- Months where the surplus is less than 855.60 are: June, July, August, September, and November.
- Conclusion for Q1: The customer’s monthly surplus is not consistently above an emi_amount of 855.60. Significant months show a deficit, demonstrating that the customer would not reliably cover an EMI of that size every month.

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2. Q2: Ability to Pay EMI Every Month Without Struggle

- Average Monthly Surplus: 855.60
- However, the data shows wide fluctuations. While May, October, and December have strong surpluses, five of the eight months are in deficit.
- Hence, even though the arithmetic average (855.60) might seem sufficient, the volatility (with deficits in June, July, August, September, and November) suggests that the customer would face stress in meeting a fixed EMI obligation every month.

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3. Q3: Stability of Income Versus Expenses

- Credit (Income) Transactions – From the Monthly Credit Table:
  – May 2023 records 8176.37
  – October 2023 is 3554.25
  – December 2023 is 6785.48
  – Several months (June, July, August, September, November) have a reported credit of 0 in the monthly summary.
- Detailed credit transaction data shows “Monthly Salary” receipts occur on specific dates. Note that while the “Monthly Salary” is a predictable income when recorded, the spread across months is irregular in the summary table.

- Debit (Expense) Transactions – The detailed list shows recurring items such as water bills, electricity bills, and utility-type payments (e.g., gym membership) along with a few one-time or irregular expenses.
- Conclusion for Q3: Although salary deposits are predictable when they occur, they do not appear every month in the summary. Expenses include many recurring items, yet the imbalance in the timing of income versus outgoes contributes to the observed volatility.

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4. Q4: Risk of Default with EMI = emi_amount

- With an assumed EMI of 855.60, five out of eight months (June, July, August, September, November) show deficits.

- The most negative surplus occurs in November (–2238.29).
- This indicates a high risk—the customer's liquidity is insufficient in several months to cover an EMI of 855.60, exposing the customer to potential default in lean months.

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5. Q5: Maximum Safe EMI Amount

- A common conservative approach is to base the safe EMI on the lowest surplus in months when income is received.
  – However, using the overall minimum monthly surplus from the table is problematic because the lowest reported (–2238.29) is negative.
- Instead, we review the positive surplus months:
  – May: 7170.56, October: 1466.27, December: 5173.79
  – Among these, October's surplus of 1466.27 is the lowest positive amount.
- To build in a safety buffer against volatility, one might choose a factor (for example, 50% of the smallest positive surplus) so that even in less favorable months the EMI burden remains manageable.
  – 50% of 1466.27 is approximately 733.
- Recommendation for Q5: A safe maximum EMI amount would be about 733 per month. This lower figure provides a buffer for months with lower or negative surpluses.

Final Recommendations:

- Suggested Safe EMI: Approximately 733 per month
  – This figure is chosen based on the smallest positive surplus (October's 1466.27) with an applied safety buffer of 50%.

- Risk Rating: High
  – Given that more than half of the months (June, July, August, September, November) register negative surpluses, the risk of failing to meet monthly EMI obligations is significant.

- Key Financial Observations:
  – There is significant variability in monthly surpluses despite sizeable income inflows in

certain months.

– The timing of salary deposits is irregular, contributing to months with negative cash flows.

– Recurring expenses (utilities and minor recurring EMIs) create obligations that cannot be met in lean months if a high EMI is imposed.

– A conservative lending approach should be adopted, ensuring that loan terms account for months when incoming cash flows are insufficient.

In conclusion, while there are periods of strong cash flow, the lack of consistent positive surplus means that any fixed EMI obligation much above 733 could expose the customer to repayment difficulties and default risk. Lending or borrowing decisions should be calibrated accordingly.

Debt-to-Income (DTI) Ratio

Below is the detailed analysis based solely on the provided tables and figures.

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1. Overall Burden

The overall Debt-to-Income (DTI) ratio is 0.0537 (approximately 5.37%). This very low ratio indicates that the borrower’s debt burden is minimal relative to income. In professional terms, a DTI of under 0.2 is generally seen as safe, and our figure is well below that benchmark.

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2. Monthly Trends

Month-to-month, only December shows a computed DTI (0.0434). The other months have missing (“nan”) DTI or incomplete Debt/Income data. For example:

- May 2023 – Income reported (8176.37), but Debt is not recorded.
- July 2023 – Debt is provided (438.1) whereas Income is missing.
- October 2023 – Income is provided (3554.25) but Debt remains absent.

December’s record with a DTI of 0.0434 is consistent with an overall low DTI (0.0537). With the incomplete data from other periods, there is no evidence of any month reaching a high-risk level. However, the missing data itself is a cautionary note, meaning that a complete record is essential for full assurance.

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3. Income & Expense Dynamics

Credit transactions indicate recurring salary deposits (e.g., May with 4850.28 and 3326.09; October with 3554.25; December with 4595.55 and 2189.93). This recurring inflow supports certainty in income.

Debit transactions, on the other hand, include a mix of recurring utility payments (e.g., water bills, electricity bills, gym memberships) and discretionary spending (e.g., movie, clothing, travel). Additionally, periodic Loan EMI payments are evident (e.g., July EMI transactions of 17.61 and 420.49, September's 262.66, and December's 294.39). Despite the diversity of spending, the relative magnitude of these debits compared with the recurring income suggests a comfortable buffer.

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4. Risk Tiering

Using common financial thresholds—0.2, 0.36, 0.43, 0.5—the borrower's overall DTI ratio of 0.0537 is dramatically below even the most conservative threshold (0.2). Therefore, from a risk perspective, the current profile would be classified in a very low-risk tier.

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5. Stability & Volatility

While full monthly DTI trends cannot be established because of missing values, the available data (with December's DTI at 0.0434 and an overall DTI of 0.0537) suggests stability. There is no evidence of a rising or volatile trend. A stable stock of recurring income combined with consistent utility and EMI outflows in the data supports a view of stable financial management.

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6. Potential Red Flags

Two issues draw attention:

- Missing Data: Several months have "nan" for either Debt or Income. For example, May's

Debt and July’s Income are missing. Although this may be due to reporting gaps rather than actual financial trouble, it is crucial to ensure complete records for accurate risk assessment.

- Data Consistency: No sudden spikes in EMI amounts are noted; however, some discretionary spending (e.g., electronics purchases, movies, travel) appears alongside regular expenses. While not immediate red flags given the low DTI, they warrant monitoring to avoid future cash flow issues.

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7. Stress Testing

If we assume that in the most recent month (December) the income drops by 10%, then the income would decrease from 6785.48 to approximately 6106.93. With December debt recorded at 294.39, the revised DTI would be:

New DTI = 294.39 / 6106.93 ≈ 0.0482 (4.82%).

This change is marginal and still well below typical high-risk thresholds. Thus, even under stress, the risk tier would remain in the very low risk category, although maintaining comprehensive data is essential.

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8. Recommendations

- Complete Financial Recording: Ensure that every month’s data for both Debt and Income are properly recorded so that all monthly DTI ratios can be accurately tracked.
- Maintain Recurring Income Streams: The borrower benefits from regular salary payments. Continued stability here underpins the low risk profile.
- Monitor Discretionary Spending: Keep discretionary outlays (e.g., movies, electronics, travel) in check. Although current ratios are healthy, these expenses can grow and impact future cash flow if not managed.
- Regular Review of EMI Payments: Ensure that all debt repayment transactions (Loan EMIs) are consistently monitored to avoid any potential accumulation of untracked debt.
- Prepare for Income Variability: Although a 10% drop does not significantly worsen the DTI at present, plan for possible reductions in income by building or maintaining an emergency cushion.

Conclusion

Based solely on the provided data, the borrower’s DTI ratio and income-to-expense dynamics suggest a very healthy financial profile with low repayment risk. Accurate and complete monthly reporting, coupled with controlled discretionary spending, will further ensure ongoing financial stability and resilience against minor income fluctuations.

Behavioral Insights

Below is the analysis based on the provided credit and debit transaction data:

1. Bounced Cheques
  - Analysis: There are no records or indicators (e.g., “bounced cheque”, “returned cheque”) in either the credit or debit datasets.
  - Risk Flag: Low
  - Contributing Factors: All transactions appear to be processed normally.

2. Speculative Transactions
  - Analysis: No transactions suggest speculative behavior. There are no entries related to stock trading, forex, or other high-risk investment instruments.
  - Risk Flag: Low
  - Contributing Factors: All transactions are either recurring (salary, bills, loan EMIs) or one-time discretionary purchases, with no evidence of market speculation.

3. Gambling
  - Analysis: None of the narratives in the datasets indicate any association with gambling merchants, casinos, betting apps, or lottery activities.
  - Risk Flag: Low
  - Contributing Factors: Even discretionary expenses (such as movies or travel) remain within

typical non-gambling behavior, with no gambling-specific terminology.

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4. Frequent Overdrafts
  - Analysis: There is no evidence of recurring overdraft fees or negative balance indications. All transactions fall within expected bill payments, purchases, and income credits.
  - Risk Flag: Low
  - Contributing Factors: Consistent income and regular payments reflect stable cash flow management, with no signs of financial distress or overdraft reliance.

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5. Heavy Use of BNPL (Buy Now Pay Later)
  - Analysis: There are no transactions that are identified or labeled as BNPL. All discretionary expenses (e.g., electronics, clothing, travel) appear to be paid in full and are not connected to deferred payment services.
  - Risk Flag: Low
  - Contributing Factors: Lack of BNPL-related nomenclature and predictable utility and EMI payments suggest disciplined spending habits.

Summary Risk Profile
  - The borrower’s financial behavior as shown in the data is predominantly stable and routine. Regular salary deposits, scheduled payments, and non-speculative discretionary spending indicate a low-risk financial profile.
  - No red flags were observed regarding bounced cheques, speculative transactions, gambling, frequent overdrafts, or heavy BNPL usage.

Overall Behavior Risk Indicator
  - Overall Score: 1 out of 10 (where 10 indicates the highest risk)

Conclusion:
Based on the transaction history provided, the borrower demonstrates disciplined and routine financial behavior with no indicators of high-risk activities. This suggests a very low overall

behavioral risk profile.

5. Risk Assessment

From a risk perspective, the primary concern lies in liquidity management. The volatility in monthly surpluses creates periods where fixed repayment commitments could lead to stress or default. This liquidity risk is evident despite an otherwise low overall debt burden and a low-risk behavioral profile. Incomplete monthly data does warrant further monitoring to confirm trends. Collectively, while behavior and DTI indicate low systemic credit risk, the irregular timing and magnitude of surpluses present a high risk if fixed high EMI obligations are imposed.

6. Conclusion & Recommendation

A conservative credit approach is advised. Lending decisions should be calibrated around a maximum safe EMI of about 733 per month to accommodate less favorable cash flow months. It is also recommended that the customer improve the consistency of monthly cash inflows and that complete financial records be maintained to better monitor liquidity trends. Overall, while the borrower is financially disciplined with low DTI, the inconsistencies in surplus flow warrant cautious credit exposure.

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Date: 2025-08-25